MILITARY SPECIFICATION SHEET
SYNCHRO CONTROL TRANSMITTER, TYPE 31CX6a

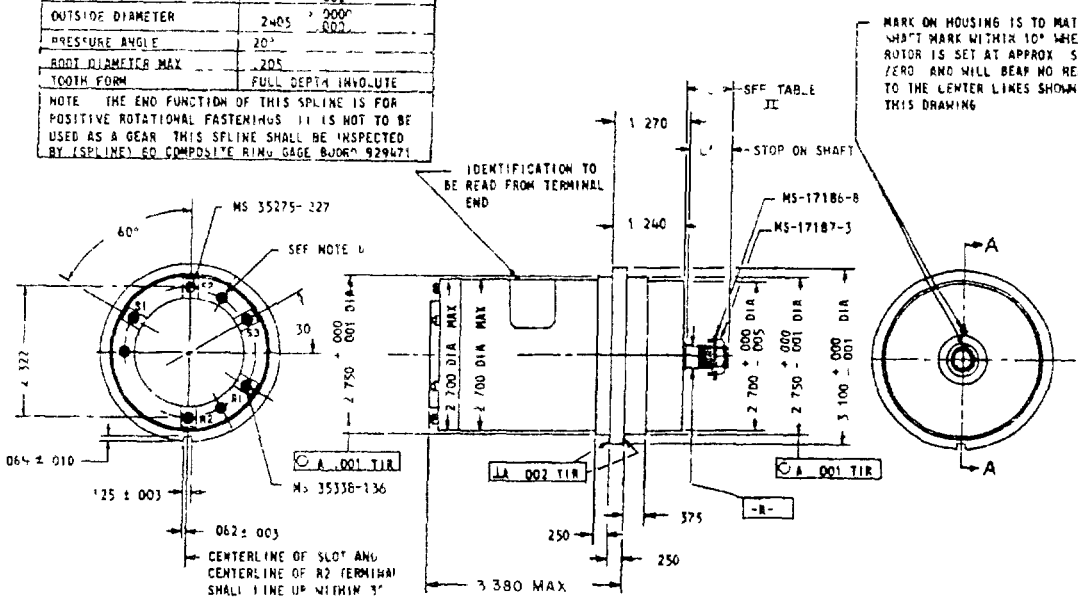
This Specification has been Approved by the Ordnance Systems Command Department of the Navy

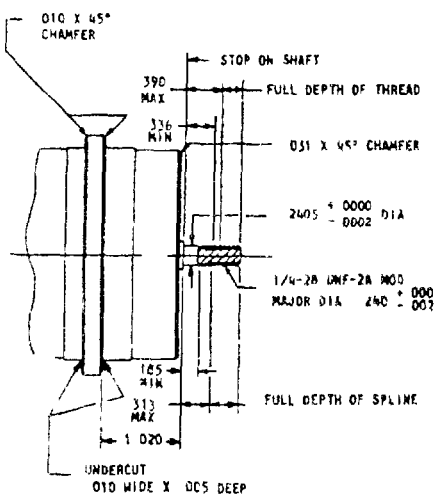
THE COMPLETE REQUIREMENTS FOR PROCURING THE SYNCHRO DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-20708

SPLINE DATA	
TEETH	22
PITCH	96
PITCH DIAMETER	2291 +.000 -.002
OUTSIDE DIAMETER	2405 +.0000 -.0002
PRESSURE ANGLE	20°
ROOT DIAMETER MAX	.205
TOOTH FORM	FULL DEPTH INVOLUTE

NOTE THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS IT IS NOT TO BE USED AS A GEAR THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE BUORD 929471

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ±1/64
DECIMALS ± 005 ANGLES ±2°





PARTIAL SECTION
A A

NOTES:

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT THEN THE HOUSING ROTATED
2. RUNOUT OF FEATURE "A" (SMOOTH PORTION) OF SHAFT SHALL NOT EXCEED .0008 TIR
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS
   END PLAY - 1 POUND RADIAL PLAY 1 POUND
4. THE TERMINAL BLOCK SHALL BE SECURED BY SCREWS OTHER THAN THE TERMINAL SCREWS RECESSES SHALL ALLOW TERMINAL LUG MS-25036-6 TO BE INSTALLED WITH THE BARRELS INBOARD OR OUTBOARD AND PREVENT THE LUGS FROM TURNING
5. MINOR VARIATIONS OF UNIT CONFIGURATION ARE PERMITTED FOR UNDIMENSIONED DESIGN DETAIL
6. THE FOLLOWING LIST OF HARDWARE MS-35275-227 MACHINE SCREW (5 REQUIRED) MS-35338-136 LOCK WASHER (5 REQUIRED) MS-17186-8 DRIVE WASHER (1 REQUIRED) MS-17187-3 NUT (1 REQUIRED) SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SYNCHRO PACKAGE FOR INFORMATION ONLY, SEE SYNCHRO CLAMP MS-17183.
7. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE SYNCHROS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN MIL-S-20708

FSC 5990

Source: https://assist.dla.mil -- Downloaded: 202640-19T08:24Z
Check the source to verify that this is the current version before use.

TABLE I

Requirements	Value	Unit	Tolerance
Frequency	60	Hz	±1%
Primary Voltage	115	volts	±1%
Primary Current	.191	milliamperes	maximum
Primary Power	2.26	watts	nominal
Impedance:			
Zro	693	ohms	±15%
Zss	63	ohms	nominal
Impedance Angle:			
Zro	83.2	degrees	±2°
Zss	20	degrees	nominal
Transformation Ratio	.783	------	±2%
Phase Shift	4.4(lead)	degrees	±.5
D.C. Resistance:			
Rotor (R1-R2)	53.6	ohms	maximum
Stator (line-to-line)	52	ohms	nominal
Electrical Error	8	minutes	maximum
Null Voltages:			
Total	100	millivolts	maximum
Fundamental	30	millivolts	maximum
Friction Torque	0.5	oz-in	maximum
Radial Play	.0008	inches	maximum
End Play	0.0030	inches	±.0020
Temperature Rise	10	degrees(C)	maximum

This unit shall not be subjected to the altitude requirements of MIL-S-20708.

Part Number: M20708/139 plus applicable dash number from Table II.

TABLE II

DASH NO.	SHAFT DIAMETER	SHAFT STYLE	TERMINAL END	L' DIMENSION TO STOP ON SHAFT	L REFERENCE FREE SHAFT LENGTH
*-001	.2405	Splined	Terminal Board	.700 ± .010	.730 ± .010

*Dash number -001 shall be the NATO Standard.

Custodians: Navy-AS

Review Activities: Navy-SH,OS

User Activities: Marine Corp. MC; U.S Coast Guard-CG

Preparing Activity: Navy-OS

Project No. 5990-N-043

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*

SPECIFICATION

MIL-S-20708/139(OS) SYNCHRO CONTROL TRANSMITTER, TYPE 31CX6a

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. PARAGRAPH NUMBER AND WORDING

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

FORM 1426
APR 61

0-8376

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Ordnance Systems Command
Washington, D.C 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL ORDNANCE SYSTEMS COMMAND (ORD 9343)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:24Z
Check the source to verify that this is the current version before use.